

Ex. No: 8 Deadlock Detection Algorithm

C PROGRAM

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
int allocation[10][10], request[10][10];
```

```
int available[10];
```

```
int work[10];
```

```
int finish[10];
```

```
int n, m;
```

```
int i, j, k;
```

```
int found;
```

```
printf("Enter Number of Processes: ");
```

```
scanf("%d", &n);
```

```
printf("Enter Number of Resource Types: ");
```

```
scanf("%d", &m);
```

```
printf("
```

```
\nEnter Allocation Matrix:
```

```
\n");
```

```
for(i = 0; i < n; i++)
```

```
{
```

```
for(j = 0; j < m; j++)
```

```
{
```

```
scanf("%d", &allocation[i][j]);
```

```
}
```

```
}  
  
printf("  
\nEnter Request Matrix:  
\n");  
  
for(i = 0; i < n; i++)  
{  
    for(j = 0; j < m; j++)  
    {  
        scanf("%d", &request[i][j]);  
    }  
}  
  
printf("  
\nEnter Available Resources:  
\n");  
  
for(i = 0; i < m; i++)  
{  
    scanf("%d", &available[i]);  
    work[i] = available[i];  
}  
  
for(i = 0; i < n; i++)  
{  
    finish[i] = 0;  
}  
  
do
```

```
{  
found = 0;  
for(i = 0; i < n; i++)  
{  
if(finish[i] == 0)
```

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```
{  
for(j = 0; j < m; j++)  
{  
if(request[i][j] > work[j])  
break;  
}  
if(j == m)  
{  
for(k = 0; k < m; k++)  
  
{  
  
work[k] += allocation[i][k];  
  
}  
  
finish[i] = 1;  
found = 1;
```

```

}
}
}
} while(found);
found = 0;
printf(
"

\nDeadlocked Processes:

\n");

for(i = 0; i < n; i++)
{
if(finish[i] == 0)
{
printf("P%d ", i);
found = 1;
}
}
if(found == 0)
{
printf("No Deadlock Detected");
}
printf("

\n");
return 0; }

```

Output:

```
(dhanush@dhanush)-[~]  
$ ./deadlockdetection  
Enter Number of Processes: 2  
Enter Number of Resource Types: 2  
  
Enter Allocation Matrix:  
12  
12  
12  
12  
  
Enter Request Matrix:  
12  
12  
12  
12  
  
Enter Available Resources:  
12  
12  
  
Deadlocked Processes:  
No Deadlock Detected
```

SHELL SCRIPT

```
#!/bin/bash  
  
echo "Enter number of processes:"  
  
read n  
  
echo "Enter deadlocked process numbers (if any):"  
  
read processes  
  
if [ -z "$processes" ]  
then  
echo "No Deadlock Detected"  
else
```

```
echo "Deadlocked Processes:"
```

```
echo "$processes"
```

```
fi
```

Output:

```
(dhanush@dhanush)-[~]  
$ nano deadlockdetection.sh  
  
(dhanush@dhanush)-[~]  
$ chmod +x deadlockdetection.sh  
  
(dhanush@dhanush)-[~]  
$ bash deadlockdetection.sh  
Enter number of processes:  
2  
Enter deadlocked process numbers (if any):  
2  
Deadlocked Processes:  
2
```